

Def.

Given four complex numbers $a, b, c, d \in \mathbb{C}$ with $ad - bc \neq 0$, a map

$$w = T(z) = \frac{az + b}{cz + d}$$

is called a linear fractional transformation.

Observation for the linear fraction.

The condition $ad - bc \neq 0$ gives that the $T(z)$ is non-constant, assume $ad - bc = 0$.

If $c \neq 0$, then

$$\frac{az + b}{cz + d} = \frac{\frac{a}{c} \cdot (cz + d) + b - \frac{ad}{c}}{cz + d} = \frac{a}{c}.$$

If $a \neq 0$, then

$$\frac{az + b}{cz + d} = \frac{az + b}{cz + \frac{bc}{a}} = \frac{a}{c} \cdot \left(\frac{z + \frac{b}{a}}{z + \frac{bc}{a^2}} \right) = \frac{a}{c} \quad \left(z \neq -\frac{b}{a} \right).$$

If $a = 0$, then either $b = 0$ or $c = 0$.

If $b = 0$, then $T \equiv 0$.

If $c = 0$, then $T \equiv \frac{b}{d}$.

If $c = 0$, then either $a = 0$ or $d = 0$.

If $a = 0$, then $T \equiv \frac{b}{d}$.

If $d = 0$, then T is not defined.

Now, we assume $ad - bc \neq 0$, then on $\mathbb{C} \setminus \{-\frac{d}{c}\}$.

$$w = T(z) = \begin{cases} \frac{a}{c} - \left[\frac{ad - bc}{c^2} \right] \cdot \frac{1}{z + \frac{d}{c}} & \text{if } c \neq 0 \\ \frac{a}{d}z + \frac{b}{d} & \text{if } c = 0, \end{cases}$$

Therefore, the T is injective.

Thm.

Define $\mathbb{H} \stackrel{\text{def}}{=} \{z \in \mathbb{C} \mid \operatorname{Im} z > 0\}$. (That is, the upper plane)

A map $F: \mathbb{H} \rightarrow \mathbb{D}: z \mapsto \frac{i-z}{i+z}$ is conformal map,

and $G: \mathbb{D} \rightarrow \mathbb{H}: w \mapsto i \cdot \frac{1-w}{1+w}$ is the inverse of F .

Proof. Given $a+ib \in \mathbb{H}$ ($b > 0$),

$$\|(0,1) - (a,b)\|^2 = a^2 + (b-1)^2 < \|(0,1) - (a,b)\|^2 = a^2 + (b+1)^2.$$

Therefore for all $z \in \mathbb{H}$, $|F(z)| < 1$, so the F is well-defined.

And, given $u+iv \in \mathbb{D}$ ($u^2+v^2 < 1$),

$$\begin{aligned} \operatorname{Im}(G(w)) &= \operatorname{Re} \left[\frac{1-(u+iv)}{1+u+iv} \right] = \operatorname{Re} \left[\frac{(1-u-iv)(1+u-iv)}{(1+u)^2+v^2} \right] \\ &= \frac{1-u^2-v^2}{(1+u)^2+v^2} > 0 \end{aligned}$$

Therefore the G is also well-defined. And, these are rational of polynomials, so holomorphic. Now, given $z \in \mathbb{H}$,

$$G(F(z)) = i \cdot \frac{1 - \frac{i-z}{i+z}}{1 + \frac{i-z}{i+z}} = i \cdot \frac{i+z - i+z}{i+z + i-z} = z, \text{ and similarly } F \circ G = \operatorname{id}.$$

So, these are bijection with $G = F^{-1}$ $\square$

Observe that for a real number $x \in \mathbb{R}$,

$$F(x) = \frac{i-x}{i+x} = \frac{(i-x)/(-i+x)}{1+x^2} = \frac{1+2xi-x^2}{1+x^2} = \frac{1-x^2}{1+x^2} + i \cdot \frac{2x}{1+x^2},$$

and parametrize $x = \tan t$ ($-\frac{\pi}{2} < t < \frac{\pi}{2}$), then

$$\sin 2a = 2 \cdot \sin a \cdot \cos a = \frac{2 \tan a}{1 + \tan^2 a} \quad \text{and} \quad \cos 2a = \frac{1 - \tan^2 a}{1 + \tan^2 a}, \quad \text{so}$$

$$F(w) = \cos 2t + i \cdot \sin 2t = e^{2it}.$$

Typical Conformal map.

Affine map

$$f: \mathbb{C} \rightarrow \mathbb{C}: z \mapsto az + b \quad (a \neq 0)$$

It magnitude $|a|$, rotating $\arg a$.

Möbius transformation

$$T: \mathbb{C} \rightarrow \mathbb{C}: z \mapsto \frac{az + b}{cz + d}, \quad \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} \neq 0.$$

Cayley transform. $\mathbb{H} := \{z \in \mathbb{C} \mid \operatorname{Im} z > 0\}$, and $\mathbb{D} = \{z \in \mathbb{C} \mid |z| < 1\}$.

$$C: \mathbb{H} \rightarrow \mathbb{D}: z \mapsto \frac{z - i}{z + i}$$

$$C^{-1}: \mathbb{D} \rightarrow \mathbb{H}: w \mapsto i \cdot \frac{1 + w}{1 - w}.$$

$$B: \{z \mid \operatorname{Re} z > 0\} \rightarrow \mathbb{D}: z \mapsto \frac{1 + z}{1 - z}$$

$$\operatorname{Aut}(\mathbb{D}) = \left\{ z \mapsto e^{i\theta} \cdot \frac{z - a}{1 - \bar{a}z} \mid |a| < 1, \theta \in \mathbb{R} \right\}$$